Nitish Ravisankar Raveendran

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Education

University of Maryland, College Park
Master of Engineering in Robotics — *Expected Graduation: Spring, 2025*
Relevant Coursework: Advanced Robotics, Machine Learning, AI in Robotics, Sensor Integration, Computer Vision, Sensor Fusion

Anna University, PEC, Chennai
Bachelor of Engineering in Electronics and Communications Engineering — *July 2022*
CGPA: 8.94/10.0

Relevant Experience

Robotics Club, PEC Chennai *President* — *March 2019 - August 2022*
- Championed club initiatives that increased membership and orchestrated multiple innovative projects and events, engaging a large number of students.

Institute of Innovations, Chennai *Intern* — *May - June 2021*
- Designed and deployed IoT devices using ESP8266 and ESP-32, improving real-time data acquisition by 20%.
- Integrated cloud-based analytics, reducing latency in sensor data processing.

Projects

Bipedal Locomotion Using Reinforcement Learning — *Fall 2024*
- Implemented **TRPO**,**PPO**, and **TQC** for bipedal locomotion in simulation, optimizing efficiency and stability.
- Compared training performance across different **RL** algorithms to refine gait adaptation.

Gesture-Based Control in Assistive Technology — *Fall 2024*
- Developed a real-time hand-gesture recognition system using **Google MediaPipe** and **ROS2**.
- Enhanced TurtleBot teleoperation efficiency by 30% through **adaptive control algorithms**.

Multi-Agent Navigation with MCTS — *Fall 2024*
- Evaluated centralized vs decentralized navigation strategies in multi-agent robotic systems.
- Integrated **Monte Carlo Tree Search (MCTS)** in **ROS2** and **Gazebo**, improving path efficiency.

Lunar Light Utility Vehicle (LLUV) — *Fall 2024*
- Conceived and built a **lightweight**, **autonomous lunar rover** for site preparation and infrastructure support.
- Integrated adaptive suspension and refined wheel designs, achieving enhanced navigation efficiency on **simulated extraterrestrial terrains**.

Arduino-based Aimbot with YOLO Object Detection — *Summer 2024*
- Engineered an advanced Arduino-based aiming solution that incorporated **YOLO** technology to achieve over 95% accuracy in target recognition while streamlining the servo adjustment process, resulting in reduced setup times across projects.

Quadcopter Design and Development — *Summer 2024*
- Devised and constructed an **autonomous quadcopter** by integrating mechanical, electronic, and software components, enhancing control responsiveness.
- Implemented simulation and control systems in **Python** and **C++** to streamline flight performance.

Reinforcement Learning for Robot Pursuit and Evasion — *Spring 2024*
- Devised a reinforcement learning framework using **DQN** for training robot pairs **(evader and pursuer)** in a 3D **ROS2** and **Gazebo** environment, bolstering strategic decision-making.

Alpha - Pick and Place Mobile Manipulator — *Fall 2023*
- Innovated and crafted a **CAD-based** mobile manipulator integrated with **ROS2**, mitigating operational errors during precision pick-and-place tasks.

Technical Skills

Programming: Python, C++, MATLAB
Machine Learning: PyTorch, TensorFlow, Deep Reinforcement Learning
Robotics: ROS2, Gazebo, Computer Vision, Perception Systems
Embedded Systems: Arduino, ESP8266, ESP-32
Simulation and Algorithms: Sensor Fusion, State Estimation, Depth Estimation, Path Planning
SBC: Raspberry Pi, NIVIDIA Jetson
Tools: JAX, Docker, Git